1 Plant and Control Law

1.1 Continuous-time plant

$$\gamma(x)\,\dot{x} \;=\; f_{dx} + f_{Dx} + f_T \tag{1}$$

1.2 Discrete-time plant (zero-order hold)

$$a_x[k] \equiv \Delta t \,/\, \gamma(x[k]) \tag{2}$$

$$x_D[k] \equiv a_x[k]\, f_{Dx}[k] \tag{3}$$

$$x[k{+}1] = x[k] + a_x[k]\big(f_{dx}[k] + f_T[k]\big) + x_D[k] \tag{4}$$

1.3 Thermal force statistics

$f_T[k]$ is zero-mean Gaussian white with variance

$$\sigma^2_{f_T}[k] \;=\; \frac{4k_BT\,\gamma(x[k])}{\Delta t} \quad\Longrightarrow\quad \mathrm{Var}\big(a_x[k]\,f_T[k]\big) \;=\; 4k_BT\,a_x[k]. \tag{5}$$

1.4 Control law

$$f_{dx}[k] \;=\; \hat{a}_x^{-1}[k]\Big[\big(x_d[k{+}1] - x_d[k]\big) + (1-\lambda_c)\,\delta\hat{x}[k] - \hat{x}_D[k]\Big] \tag{6}$$

In the augmented state of §3, $\delta\hat{x}[k] = \delta\hat{x}_3[k]$ (due to the two-step measurement delay; see §3).

2 IIR Variance Estimator for $a_x[k]$

This section details the infinite impulse response (IIR) variance estimator that produces the auxiliary measurement a_{xm} from the measured tracking error. The pipeline combines low-pass (LP) and high-pass (HP) stages.

2.1 Tracking error and high-pass filter

$$\delta x_m[k] = x_d[k{-}2] - x_m[k] \tag{7}$$

$$\delta x_{md}[k] = (1-a_{pd})\,\delta x_{md}[k{-}1] + a_{pd}\,\delta x_m[k] \tag{8}$$

$$\delta x_{mr}[k] = \delta x_m[k] - \delta x_{md}[k] \tag{9}$$

Only δx_{mr} is driven by $a_x\,f_T$; δx_{md} holds the trajectory-tracking mean.

2.2 Single-sample variance estimator

$\sigma^2(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$ with two IIR sub-estimators:

$$\delta x_{mrd}[k] = (1-a_{prd})\,\delta x_{mrd}[k{-}1] + a_{prd}\,\delta x_{mr}[k] \tag{10}$$

$$\overline{(\delta x_{mr})^2}[k] = (1-a_{cov})\,\overline{(\delta x_{mr})^2}[k{-}1] + a_{cov}\,\big(\delta x_{mr}[k]\big)^2 \tag{11}$$

$$\sigma^2_{\delta x_{mr}}[k] = \overline{(\delta x_{mr})^2}[k] - \big(\delta x_{mrd}[k]\big)^2 \tag{12}$$

2.3 Role of the three IIR coefficients

coef	role	analyzed in
a_{pd}	LP pole of δx_m; defines δx_{mr}	§5 (C_{dpmr}, C_n)
a_{prd}	LP pole of δx_{mr}	§6 (β)
a_{cov}	LP pole of $\overline{(\delta x_{mr})^2}$	bias-free; estimator variance only

2.4 Link between $\sigma^2_{\delta x_{mr}}$ and a_x

$$\boxed{\sigma^2_{\delta x_{mr}}[k] \;\approx\; \beta\left(C_{\text{dpmr}} \cdot 4k_B T\, a_x[k] + C_n\, \sigma_n^2\right)} \tag{13}$$

(C_{dpmr}, C_n) derived in §5; β in §6.

$$a_{xm}[k] \;=\; \frac{\sigma^2_{\delta x_{mr}}[k]/\beta \;-\; C_n\, \sigma_n^2}{C_{\text{dpmr}} \cdot 4k_B T} \tag{14}$$

3 Kalman Filter (KF)

3.1 State vector and process model

$$\boldsymbol{x}[k] = \left[\,\delta x_1,\ \delta x_2,\ \delta x_3,\ x_D,\ \delta x_D,\ a_x,\ \delta a_x\,\right]^T[k] \tag{15}$$

$$\boldsymbol{x}[k{+}1] = \boldsymbol{F}_e\, \boldsymbol{x}[k] + \boldsymbol{q}[k] \tag{16}$$

$$\boldsymbol{F}_e = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & -1 & 0 & -f_{dx}[k] & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

3.2 Measurement model

$$\boldsymbol{y}[k] = \boldsymbol{H}\, \boldsymbol{x}[k] + \boldsymbol{v}[k] = \begin{bmatrix} \delta x_m[k] \\ a_{xm}[k] \end{bmatrix} \tag{17}$$

$$\boldsymbol{H} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

3.3 Kalman filter recursions

$$\boldsymbol{L}[k] = \boldsymbol{P}_f[k]\, \boldsymbol{H}^T \big(\boldsymbol{H}\, \boldsymbol{P}_f[k]\, \boldsymbol{H}^T + \boldsymbol{R}\big)^{-1} \tag{18}$$

$$\boldsymbol{P}[k] = (\boldsymbol{I} - \boldsymbol{L}[k]\, \boldsymbol{H})\, \boldsymbol{P}_f[k] \tag{19}$$

$$\boldsymbol{P}_f[k{+}1] = \boldsymbol{F}_e\, \boldsymbol{P}[k]\, \boldsymbol{F}_e^T + \boldsymbol{Q} \tag{20}$$

$$\hat{\boldsymbol{x}}[k{+}1\,|\,k] = \boldsymbol{F}_e\, \hat{\boldsymbol{x}}[k\,|\,k{-}1] + \boldsymbol{F}_e\, \boldsymbol{L}_{ss}\big(\boldsymbol{y}[k] - \boldsymbol{H}\, \hat{\boldsymbol{x}}[k\,|\,k{-}1]\big) \tag{21}$$

5 Closed-Loop Steady-State Variance: C_{dpmr} and C_n

Scope note. The derivation drops the time-varying entry $\boldsymbol{F}_e(3,6) = -f_{dx}[k]$ from the $\boldsymbol{A}$ matrix to obtain a time-invariant system so that Lyapunov analysis applies. Under zero-mean f_{dx} operating conditions (static regulation with $\Delta p_d = 0$, or symmetric oscillatory reference), the dropped $f_{dx} \cdot e_a$ term has $\mathrm{E}[f_{dx} \cdot e_a] \approx 0$ and contributes variance of order $\sigma_f^2 \sigma_e^2$, which is negligible relative to the dominant thermal term $4k_B T\, a_x$ (i.e. $\sigma_f^2 \sigma_e^2 / (4k_B T\, a_x) \ll 1$). Empirical variance matches the derived C_{dpmr} at all operating points tested (free-space, near-wall static, dynamic sweep).

5.1 Augmented state and closed-loop dynamics

$$\boldsymbol{e}[k] \;=\; \big[\,\delta x_1,\ \delta x_2,\ \delta x_3,\ \ e_{x1},\ e_{x2},\ e_{x3},\ \ e_{xD},\ e_{\delta xD},\ e_a,\ e_{\delta a},\ \ \delta x_{md}[k{-}1]\,\big]^T[k] \;\in\; \mathbb{R}^{11}$$

$$\boldsymbol{e}[k{+}1] \;=\; \boldsymbol{A}\,\boldsymbol{e}[k] \;+\; \boldsymbol{b}_T\big(a_x[k]\, f_T[k]\big) \;+\; \boldsymbol{b}_n\, n[k] \tag{22}$$

$$\boldsymbol{A} \;=\; \left[\begin{array}{ccc|cccccccc|c}
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \lambda_c & 0 & 0 & 1{-}\lambda_c & -1 & 0 & 0 & 0 & 0 \\
\hline
\multicolumn{3}{c|}{\boldsymbol{0}_{7\times 3}} & \multicolumn{8}{c|}{\boldsymbol{A}_e} & \boldsymbol{0}_{7\times 1} \\
\hline
a_{pd} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1{-}a_{pd}
\end{array}\right]$$

where $\boldsymbol{A}_e \equiv \boldsymbol{F}_e(\boldsymbol{I} - \boldsymbol{L}_{ss}\boldsymbol{H})$ is the steady-state KF estimation-error dynamics (§3.3).

$$\boldsymbol{b}_T \;=\; \big[\,0,\ 0,\ -1,\ \ 0,\ 0,\ -1,\ 0,\ 0,\ 0,\ 0,\ \ 0\,\big]^T, \qquad \boldsymbol{b}_n \;=\; \big[\,0,\ 0,\ 0,\ \ -\boldsymbol{F}_e\boldsymbol{L}_{ss}(:,1),\ \ -a_{pd}\,\big]^T$$

5.2 Lyapunov decomposition

Define $\boldsymbol{\Sigma}[k] \equiv \mathrm{E}\big[\boldsymbol{e}[k]\boldsymbol{e}[k]^T\big]$. Since $f_T[k]$ and $n[k]$ are independent zero-mean white processes, $\boldsymbol{\Sigma}$ decomposes linearly:

$$\boldsymbol{\Sigma}[k] \;=\; 4k_B T\, a_x[k]\, \boldsymbol{\Sigma}_T \;+\; \sigma_n^2\, \boldsymbol{\Sigma}_n \tag{23}$$

$$\boldsymbol{\Sigma}_T \;=\; \boldsymbol{A}\,\boldsymbol{\Sigma}_T\,\boldsymbol{A}^T \;+\; \boldsymbol{b}_T\,\boldsymbol{b}_T^T \tag{24}$$

$$\boldsymbol{\Sigma}_n \;=\; \boldsymbol{A}\,\boldsymbol{\Sigma}_n\,\boldsymbol{A}^T \;+\; \boldsymbol{b}_n\,\boldsymbol{b}_n^T \tag{25}$$

5.3 Variance of δx_{mr}

Substituting the §2.1 LP recursion into the HP residual definition, with $\delta x_m[k] = \delta x_1[k] - n[k]$:

$$\delta x_{mr}[k] \;=\; (1 - a_{pd})\big(\delta x_1[k] - \delta x_{md}[k{-}1] - n[k]\big) \tag{26}$$

Variance ($n[k]$ independent of $\boldsymbol{e}[k]$):

$$\sigma^2_{\delta x_{mr}}[k] \;=\; (1 - a_{pd})^2\Big[\,(\boldsymbol{\Sigma}[k])_{1,1} \;-\; 2\,(\boldsymbol{\Sigma}[k])_{1,11} \;+\; (\boldsymbol{\Sigma}[k])_{11,11} \;+\; \sigma_n^2\,\Big] \tag{27}$$

5.4 C_{dpmr} and C_n

Substituting $\boldsymbol{\Sigma}[k] \;=\; 4k_B T\, a_x[k]\, \boldsymbol{\Sigma}_T + \sigma_n^2\, \boldsymbol{\Sigma}_n$ into the §5.3 variance formula and matching against the target $\sigma^2_{\delta x_{mr}} = C_{\text{dpmr}} \cdot 4k_B T\, a_x + C_n\, \sigma_n^2$:

$\boldsymbol{\Sigma}_T$ and $\boldsymbol{\Sigma}_n$ are the discrete Lyapunov solutions — equivalently, geometric series in $\boldsymbol{A}$ — driven by $\boldsymbol{b}_T$ and $\boldsymbol{b}_n$ respectively:

$$\boldsymbol{\Sigma}_T(\lambda_c, a_{pd};\, \boldsymbol{Q}, \boldsymbol{R}) \;=\; \sum_{k=0}^{\infty} \boldsymbol{A}^k\, \boldsymbol{b}_T \boldsymbol{b}_T^T\, (\boldsymbol{A}^T)^k$$

$$\boldsymbol{\Sigma}_n(\lambda_c, a_{pd};\, \boldsymbol{Q}, \boldsymbol{R}) \;=\; \sum_{k=0}^{\infty} \boldsymbol{A}^k\, \boldsymbol{b}_n \boldsymbol{b}_n^T\, (\boldsymbol{A}^T)^k$$

(solutions of $\boldsymbol{A}\boldsymbol{\Sigma}_T\boldsymbol{A}^T + \boldsymbol{b}_T\boldsymbol{b}_T^T = \boldsymbol{\Sigma}_T$ and $\boldsymbol{A}\boldsymbol{\Sigma}_n\boldsymbol{A}^T + \boldsymbol{b}_n\boldsymbol{b}_n^T = \boldsymbol{\Sigma}_n$; $\boldsymbol{A}, \boldsymbol{b}_T, \boldsymbol{b}_n$ constructed per §5.1.)

Therefore:

$$\boxed{C_{\mathrm{dpmr}}(\lambda_c, a_{pd};\, \boldsymbol{Q}, \boldsymbol{R}) \;=\; (1-a_{pd})^2 \big[\, (\boldsymbol{\Sigma}_T)_{1,1} \;-\; 2\,(\boldsymbol{\Sigma}_T)_{1,11} \;+\; (\boldsymbol{\Sigma}_T)_{11,11} \,\big]} \tag{28}$$

$$\boxed{C_n(\lambda_c, a_{pd};\, \boldsymbol{Q}, \boldsymbol{R}) \;=\; (1-a_{pd})^2 \big[\, (\boldsymbol{\Sigma}_n)_{1,1} \;-\; 2\,(\boldsymbol{\Sigma}_n)_{1,11} \;+\; (\boldsymbol{\Sigma}_n)_{11,11} \;+\; 1 \,\big]} \tag{29}$$

The $+1$ in C_n captures the direct contribution of the current $n[k]$ via the explicit $-n$ term in §5.3; the $(\boldsymbol{\Sigma}_n)$ selector captures past $n[k]$'s propagated through the state dynamics $\boldsymbol{e}[k]$. With $(\boldsymbol{Q}, \boldsymbol{R})$ fixed at design time, the LUT is built offline on a (λ_c, a_{pd}) grid:

```
for each (λc, apd) in design grid:
  build A and bn per §5.1 from (λc, apd; Q, R, H)
  ΣT ← dlyap(A, bT bT^T)
  Σn ← dlyap(A, bn bn^T)
  Cdpmr[λc, apd] ← (1 − apd)^2 [(ΣT)1,1 − 2(ΣT)1,11 + (ΣT)11,11]
  Cn[λc, apd] ← (1 − apd)^2 [(Σn)1,1 − 2(Σn)1,11 + (Σn)11,11 + 1]
```

6 IIR Finite-Sample Bias Factor β

6.1 Bias source

The Section 2 IIR algorithm computes $\sigma^2_{\delta x_{mr}}[k] = \overline{(\delta x_{mr})^2}[k] - (\delta x_{mrd}[k])^2$. In stationary steady state (δx_{mr} zero-mean):

$$\mathrm{E}\big[\, \overline{(\delta x_{mr})^2} \,\big] \;=\; \mathrm{Var}(\delta x_{mr}) \tag{30}$$

$$\mathrm{E}\big[\, (\delta x_{mrd})^2 \,\big] \;=\; \mathrm{Var}(\delta x_{mrd}) \tag{31}$$

so the IIR output systematically underestimates $\mathrm{Var}(\delta x_{mr})$ by the amount $\mathrm{Var}(\delta x_{mrd}) > 0$. Define the bias factor β as the ratio of the IIR estimator's expected output $\mathrm{E}[\sigma^2_{\delta x_{mr}}]$ to the true variance $\mathrm{Var}(\delta x_{mr})$:

$$\beta \;\equiv\; \frac{\mathrm{E}\big[\, \sigma^2_{\delta x_{mr}} \,\big]}{\mathrm{Var}(\delta x_{mr})} \;=\; 1 \;-\; \frac{\mathrm{Var}(\delta x_{mrd})}{\mathrm{Var}(\delta x_{mr})} \tag{32}$$

6.2 $\mathrm{Var}(\delta x_{mr})$ and autocorrelation $\rho(L)$

To evaluate β, we need the denominator $\mathrm{Var}(\delta x_{mr})$ and the temporal autocorrelation $\rho(L)$ used for the numerator.

For the noise-free case ($\sigma_n^2 = 0$), §5's selector form gives

$$\mathrm{Var}(\delta x_{mr}) \;=\; (1-a_{pd})^2 \big[\, (\boldsymbol{\Sigma}_T)_{1,1} - 2(\boldsymbol{\Sigma}_T)_{1,11} + (\boldsymbol{\Sigma}_T)_{11,11} \,\big] \cdot 4k_BT\, a_x. \tag{33}$$

The lag-L autocorrelation is defined as

$$\rho(L) \;\equiv\; \mathrm{Cov}\big(\delta x_{mr}[k],\, \delta x_{mr}[k-L]\big) \,/\, \mathrm{Var}(\delta x_{mr}). \tag{34}$$

Since $\boldsymbol{A}^L\boldsymbol{\Sigma}_T$ is the lag-L cross-covariance of $\boldsymbol{e}[k]$, applying the same selector (the common $(1-a_{pd})^2$ and $4k_BT\,a_x$ prefactors cancel in the ratio) yields

$$\rho(L) \;=\; \frac{(\boldsymbol{A}^L\boldsymbol{\Sigma}_T)_{1,1} - 2(\boldsymbol{A}^L\boldsymbol{\Sigma}_T)_{1,11} + (\boldsymbol{A}^L\boldsymbol{\Sigma}_T)_{11,11}}{(\boldsymbol{\Sigma}_T)_{1,1} - 2(\boldsymbol{\Sigma}_T)_{1,11} + (\boldsymbol{\Sigma}_T)_{11,11}}. \tag{35}$$

6.3 $\mathrm{Var}(\delta x_{mrd})$ closed form

Unrolling the §2.2 LP recursion converts $\delta x_{mrd}[k]$ into an explicit weighted sum of past δx_{mr}, so that $\mathrm{Var}(\delta x_{mrd})$ can be expressed via $\rho(L)$:

$$\delta x_{mrd}[k] \;=\; a_{prd}\sum_{j=0}^{\infty}(1-a_{prd})^j\,\delta x_{mr}[k-j] \tag{36}$$

Take variance, split diagonal $j=l$ from off-diagonal $j\neq l$, and use $\sum_{j=0}^{\infty}(1-a_{prd})^{2j} = 1/\big[a_{prd}(2-a_{prd})\big]$:

$$\mathrm{Var}(\delta x_{mrd}) \;=\; \frac{a_{prd}}{2-a_{prd}}\,\mathrm{Var}(\delta x_{mr})\,\Big[\,1 + 2\sum_{m=1}^{\infty}\rho(m)(1-a_{prd})^m\,\Big] \tag{37}$$

6.4 Bias factor β

Substituting the $\mathrm{Var}(\delta x_{mrd})$ expression into the β definition:

$$\boxed{\beta(\lambda_c, a_{pd}, a_{prd};\, \boldsymbol{Q}, \boldsymbol{R}) \;=\; 1 \;-\; \frac{a_{prd}}{2-a_{prd}}\,\Big[\,1 + 2\sum_{m=1}^{\infty}\rho(m)(1-a_{prd})^m\,\Big]} \tag{38}$$

The bracket is the *autocorrelation amplification*: $=1$ for white δx_{mr}, >1 for positively-correlated.

7 Q/R Design

This section derives each non-zero entry of the Kalman filter (KF) process-noise $\boldsymbol{Q}$ and measurement-noise $\boldsymbol{R}$ from physical first principles.

7.1 Zero elements of $\boldsymbol{Q}$

$Q(i,i)$ represents the process noise entering state i. Under *positioning* (probe held at fixed h_{init}), most states have no driver:

- $\delta x_1, \delta x_2$: past values of δx_3, shifted by delay.
- $x_D, \delta x_D$: disturbance is modeled as zero ($x_D \equiv 0$).
- δa_x: $a = a_{\mathrm{nom}}/c_{\mathrm{axis}}(\bar{h})$ is constant at fixed h, so $\delta a \equiv 0$.

Only two entries are non-zero: $Q(3,3)$ (thermal force injection into δx_3) and $Q(6,6)$ (a small positive regularization value, see below).

7.2 Non-zero elements of Q

$Q(3,3)$ — **thermal force injection.**

Per-axis thermal-force variance from the Einstein relation with wall-corrected drag:

$$\mathrm{Var}(f_{T,\mathrm{axis}}) \;=\; \frac{4k_BT\,\gamma_N\,c_{\mathrm{axis}}(\bar{h})}{\Delta t}. \tag{39}$$

Multiplying by a^2_{axis} gives the thermal injection into δx_3:

$$\mathrm{Var}(a_{\mathrm{axis}}\,f_T) \;=\; 4k_BT\cdot a_{\mathrm{axis}} \;=\; a_{\mathrm{ratio}}\cdot\sigma^2_{\mathrm{dXT}}, \tag{40}$$

with $\sigma^2_{\mathrm{dXT}} \equiv 4k_BT\,a_{\mathrm{nom}}$, $a_{\mathrm{nom}} \equiv \Delta t/\gamma_N$, $a_{\mathrm{ratio}} \equiv a_{\mathrm{axis}}/a_{\mathrm{nom}}$. Hence

$$\boxed{Q(3,3) \;=\; a_{\mathrm{ratio}}\cdot\sigma^2_{\mathrm{dXT}}} \tag{41}$$

$Q(6,6)$ — **a small positive value ϵ.**

Under positioning, a is constant, so physically $Q(6,6) = 0$. But setting $Q(6,6) = 0$ together with $Q(7,7) = 0$ makes the filter assume a never varies: the discrete algebraic Riccati equation (DARE) then solves to $\boldsymbol{L}_{ss}(6,:) = 0$, meaning the filter's gain on the a_{xm} measurement is zero — a_{xm} gets ignored entirely. To prevent this, add a small positive floor:

$$Q(6,6) \;=\; \epsilon\cdot\sigma^2_{\mathrm{dXT}}, \qquad \epsilon \ll 1. \tag{42}$$

This tells the filter to allow a tiny slow drift in a, restoring a small non-zero $\boldsymbol{L}_{ss}(6,:)$ so that a_{xm} can slowly update $\hat{a}$. ϵ is chosen small enough to preserve the scalar Riccati approximation used in §8.3.

7.3 Measurement noise R (per axis)

$R(1,1)$ — **sensor noise on δx_m.** The position measurement has per-axis sensor-noise variance $\sigma^2_{n,\mathrm{axis}}$. Normalizing by the thermal unit:

$$\boxed{R(1,1)_{\mathrm{axis}} \;=\; \sigma^2_{n,\mathrm{axis}}\,/\,\sigma^2_{\mathrm{dXT}}} \tag{43}$$

Per-axis is essential because different axes' tracking algorithms have σ_n values spanning orders of magnitude.

$R(2,2)$ — **sampling noise on a_{xm}.**

a_{xm} is not a direct measurement — it is the output of the IIR variance estimator (§2.4). Its finite-sample noise variance takes the form

$$\mathrm{Var}(n_a) \;=\; \chi_{\mathrm{sq}}\cdot\rho_a\cdot a^2_{\mathrm{axis}}, \tag{44}$$

built from three factors:

- $\chi_{\mathrm{sq}} \equiv 2a_{cov}/(2-a_{cov})$: exponential moving average (EMA) finite-sample factor.
- $\rho_a \equiv 1 + 2\sum_{L\geq 1}\rho^2(L)\,(1-a_{cov})^L$: autocorrelation amplification factor (≥ 1).
- a^2_{axis}: signal scale (a_{xm} scales with a).

The squared ρ^2 (vs. §6.4's linear ρ) arises because the EMA acts on $(\delta x_{mr})^2$: Wick/Isserlis gives the lag-L autocorrelation of the squared Gaussian process as $\rho^2(L)$.

The $\rho(L)$ used here has the same selector structure as §6.2 but with $V_{\mathrm{state}} + \sigma^2_n$ in the denominator (§6.2's form is the $\sigma^2_n = 0$ limit).

Normalizing by the thermal unit:

$$\boxed{R(2,2)_{\mathrm{axis}} \;=\; \chi_{\mathrm{sq}}\cdot\rho_{a,\mathrm{axis}}\cdot a^2_{\mathrm{axis}}\,/\,\sigma^2_{\mathrm{dXT}}} \tag{45}$$

7.4 Summary: Q and R as functions of inputs

$$\boxed{Q(3,3)(\bar{h}) \;=\; a_{\text{ratio}}(\bar{h}) \cdot \sigma^2_{\text{dXT}}} \tag{46}$$

$$\boxed{Q(6,6)(\epsilon) \;=\; \epsilon \cdot \sigma^2_{\text{dXT}}} \tag{47}$$

$$\boxed{R(1,1)_{\text{axis}}\big(\sigma^2_{n,\text{axis}}\big) \;=\; \sigma^2_{n,\text{axis}} \,/\, \sigma^2_{\text{dXT}}} \tag{48}$$

$$\boxed{R(2,2)_{\text{axis}}\big(\bar{h},\, a_{cov},\, \rho_{a,\text{axis}}\big) \;=\; \chi_{\text{sq}} \cdot \rho_{a,\text{axis}} \cdot a^2_{\text{axis}}(\bar{h}) \,/\, \sigma^2_{\text{dXT}}} \tag{49}$$

Note on $\rho_{a,\text{axis}}$ dependency: via $\boldsymbol{A}_{\text{aug}}$ and $\boldsymbol{\Sigma}_{\text{aug}}$, $\rho_{a,\text{axis}}$ itself is a function of (λ_c, a_{pd}, a_{cov}, $\bar{h}$, ϵ, $\sigma^2_{n,\text{axis}}$) via a self-consistent fixed point. Hence $R(2,2)_{\text{axis}}$ ultimately depends on the same input set.

8 Theoretical Performance

Combining the Q/R design of §7 with the augmented Lyapunov framework of §5 gives closed-form predictions for the tracking-error variance and the motion-gain estimation variance per axis.

8.1 Three-channel augmented covariance

The two-channel decomposition of §5.2 is extended with a third channel for the a_{xm} sampling noise. Since f_T, n_p, and n_a are mutually independent white noises, the augmented covariance splits linearly:

$$\boxed{\boldsymbol{\Sigma}_{\text{aug}} \;=\; a_{\text{ratio}}\, \sigma^2_{\text{dXT}}\, \boldsymbol{\Sigma}_T \;+\; \sigma^2_n\, \boldsymbol{\Sigma}_n \;+\; \chi_{\text{sq}}\, \rho_a\, a^2\, \boldsymbol{\Sigma}_{na}} \tag{50}$$

Each $\boldsymbol{\Sigma}_X$ solves the unit-variance Lyapunov equation $\boldsymbol{\Sigma}_X = \boldsymbol{A}\,\boldsymbol{\Sigma}_X\,\boldsymbol{A}^T + \boldsymbol{b}_X\,\boldsymbol{b}_X^T$ with input vector

$$\begin{aligned}
\boldsymbol{b}_T &= \big[0,0,-1;\; 0,0,-1,0,0,0,0;\; 0\big]^T && \text{(thermal; §5.1)}\\
\boldsymbol{b}_n &= \big[0,0,0;\; -\boldsymbol{F}_e\boldsymbol{L}_{ss}(:,1);\; -a_{pd}\big]^T && \text{(sensor; §5.1)}\\
\boldsymbol{b}_{na} &= \big[0,0,0;\; -\boldsymbol{F}_e\boldsymbol{L}_{ss}(:,2);\; 0\big]^T && \text{(new: } a_{xm} \text{ channel)}
\end{aligned}$$

$\boldsymbol{b}_{na}$ mirrors $\boldsymbol{b}_n$ but uses the second column of $\boldsymbol{L}_{ss}$ (corresponding to the a_{xm} measurement) and has a zero IIR entry because a_{xm} does not feed the δx_{md} filter.

8.2 Tracking-error variance (per axis)

The first diagonal element of $\boldsymbol{\Sigma}_{\text{aug}}$ (corresponding to δx_1, the tracking error) gives

$$\boxed{\text{Var}(\delta x_{\text{axis}}) \;=\; (\boldsymbol{\Sigma}_{\text{aug}})_{1,1}} \tag{51}$$

with three additive contributions:

- thermal: $a_{\text{ratio}}\, \sigma^2_{\text{dXT}}\, (\boldsymbol{\Sigma}_T)_{1,1}$
- sensor: $\sigma^2_n\, (\boldsymbol{\Sigma}_n)_{1,1}$
- a_{xm} sampling: $\chi_{\text{sq}}\, \rho_a\, a^2\, (\boldsymbol{\Sigma}_{na})_{1,1}$

Per-axis scaling enters through a_{ratio}, $\sigma^2_{n,\text{axis}}$, a_{axis}, and $\rho_{a,\text{axis}}$.

8.3 Motion-gain estimation (per axis)

We evaluate the *coefficient of variation* (CV) of the motion-gain estimate per axis:

$$\mathrm{CV}(\hat{a}_{\mathrm{axis}}) \;\equiv\; \mathrm{std}(\hat{a}_{\mathrm{axis}} - a_{\mathrm{axis}}) \,/\, a_{\mathrm{axis}}. \tag{52}$$

CV^2 is additive across independent noise sources. Two sources contribute here:

- **KF estimation error**: the KF cannot perfectly track a from noisy a_m.
- **Wall-effect on a_{true}**: $a_{\mathrm{true}}(\bar{h})$ itself varies as $\bar{h}$ fluctuates with the probe's z-error δp_z; independent of KF.

(Here c_{axis} denotes $c_{\parallel}$ for x, y axes or $c_{\perp}$ for the z axis, as defined in §7.2.) Thus

$$\mathrm{CV}^2(\hat{a}_{\mathrm{axis}}) \;=\; \mathrm{CV}^2_{\mathrm{KF}} \;+\; \mathrm{CV}^2_{\mathrm{wall}}. \tag{53}$$

Block A: KF estimation term.

Step 1 — Decoupling. $Q(7,7) = 0$ (§7.1) decouples the $(a, \delta a)$ KF block; the a-subsystem reduces to a scalar KF tracking a from noisy a_m.

Step 2 — Scalar Riccati gain. The scalar discrete algebraic Riccati equation yields

$$L_{\mathrm{eff,\,axis}} \;=\; \sqrt{Q(6,6) \,/\, R(2,2)_{\mathrm{axis}}}. \tag{54}$$

(L_{eff} matches the full DARE gain $\boldsymbol{L}_{ss}(6,2)$ when $Q(7,7) = 0$; the approximation breaks at $Q(7,7) > 0$.)

Step 3 — Input noise characterization. The KF sees $a_m = a + n_a$ with $\mathrm{Var}(n_a) = \chi_{\mathrm{sq}}\, \rho_a\, a^2_{\mathrm{axis}}$ (§7.3). Importantly, n_a is a first-order autoregressive process (AR(1)) with bandwidth a_{cov}, inherited from the IIR EMA.

Step 4 — Cascaded LP. The KF (bandwidth L_{eff}) low-passes the AR(1) noise n_a (bandwidth a_{cov}). A two-state variance analysis (equivalent to integrating the cascaded AR(1) transfer function in the frequency domain) gives, in the small-L_{eff} and small-a_{cov} limit:

$$\mathrm{Var}(\hat{a} - a) \,/\, \mathrm{Var}(n_a) \;=\; L_{\mathrm{eff}} \,/\, (L_{\mathrm{eff}} + a_{cov}). \tag{55}$$

This replaces the standard $L_{\mathrm{eff}}/2$ result that would apply if n_a were white noise.

Combining Step 3 and Step 4:

$$\boxed{\mathrm{CV}^2_{\mathrm{KF,\,axis}} \;=\; \chi_{\mathrm{sq}} \cdot \rho_{a,\mathrm{axis}} \cdot \frac{L_{\mathrm{eff,\,axis}}}{L_{\mathrm{eff,\,axis}} + a_{cov}}} \tag{56}$$

Block B: Wall-effect on a_{true}.

$a_{\mathrm{true}}(\bar{h}) = a_{\mathrm{nom}}/c_{\mathrm{axis}}(\bar{h})$ depends on wall distance. Since $\delta\bar{h} = \delta p_z/R$ (up to sign), linearizing gives:

$$\boxed{\mathrm{CV}^2_{\mathrm{wall,\,axis}} \;=\; \left(\frac{|dc_{\mathrm{axis}}/d\bar{h}|}{c_{\mathrm{axis}}}\right)^2 \cdot \frac{\mathrm{Var}(\delta p_z)}{R^2}} \tag{57}$$

Total (assuming independence):

$$\boxed{\mathrm{CV}^2(\hat{a}_{\mathrm{axis}}) \;=\; \chi_{\mathrm{sq}}\, \rho_{a,\mathrm{axis}} \frac{L_{\mathrm{eff}}}{L_{\mathrm{eff}} + a_{cov}} \;+\; \left(\frac{|dc_{\mathrm{axis}}/d\bar{h}|}{c_{\mathrm{axis}}}\right)^2 \frac{\mathrm{Var}(\delta p_z)}{R^2}} \tag{58}$$

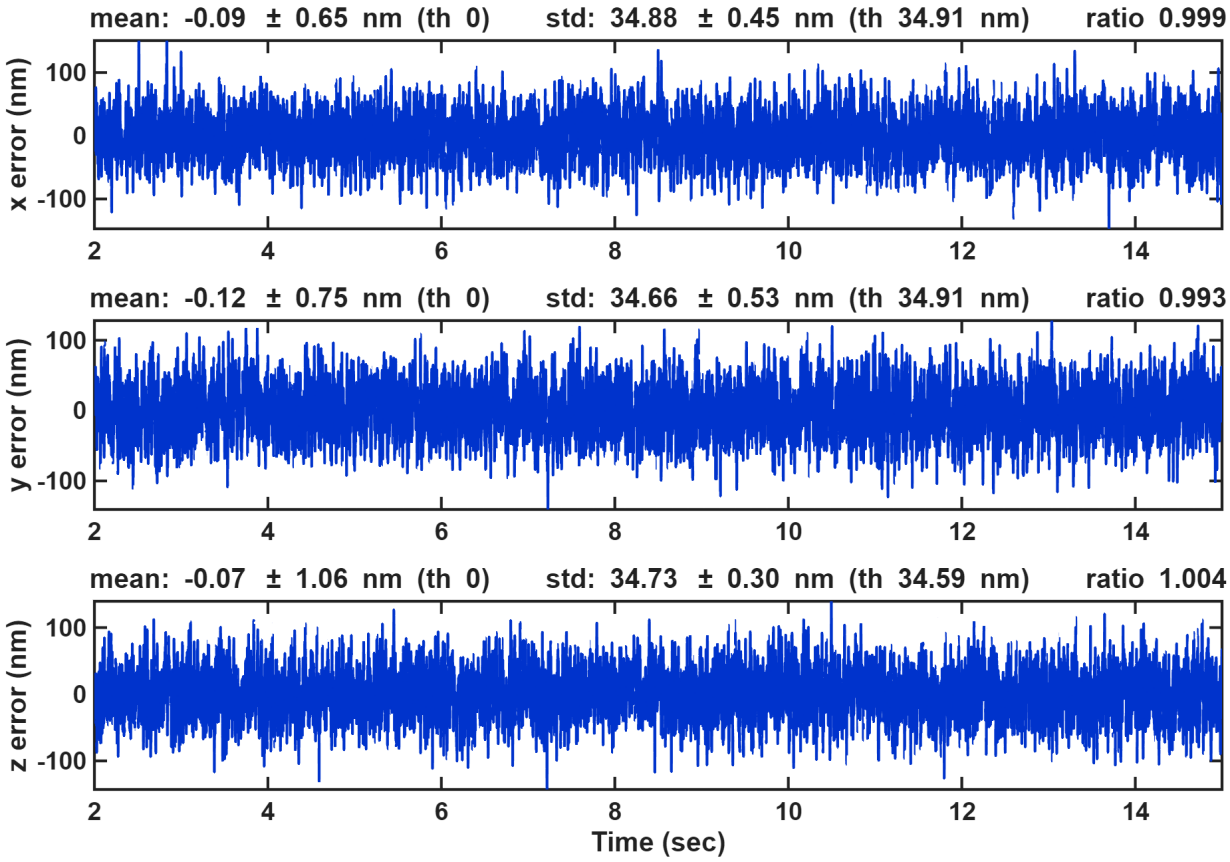

Figure 1: Tracking error per axis ($h = 50\,\mu$m).

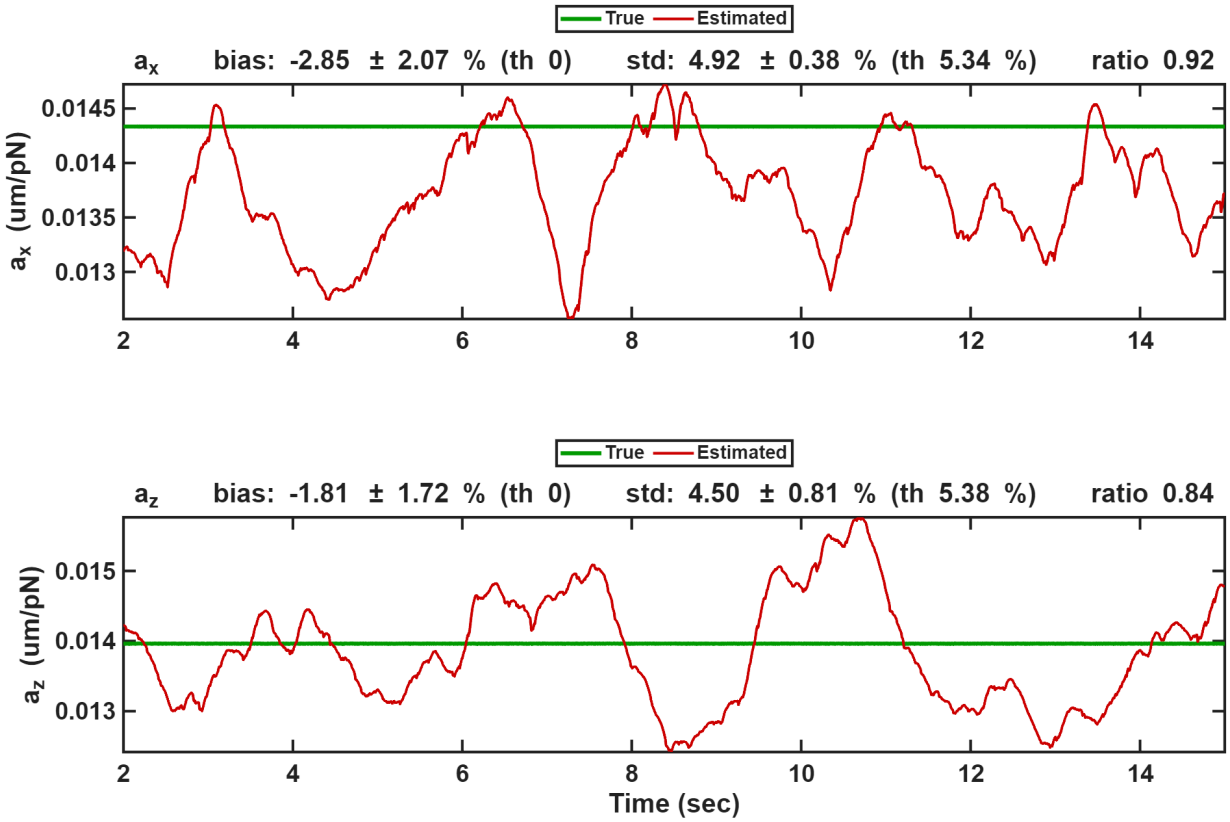

Figure 2: Motion-gain estimation ($h = 50\,\mu$m).